

Software Design Brief: Mental Health App

Introduction

This is a proposition for a software design brief, concerned about the development for a mental health app designed for said mental health charity. The charity seeks to build an app catered to the mental healthcare for university students during their university life. The charity has a particular interest in the ideas of good sleep, food, exercise and possibly many more ideas related to the wellbeing of the student in their university life. The creation of the brief has taken into account the desire for highly ambitious, creative and daring projects, while utilising pre-existing technologies. These include the examples of intricate smartphone functionalities, and deep-rooted collaboration with many respected universities and institutions.

The ideas generated from this report was carefully generated from extensive research into the student community, primarily via interviews. The findings resulted in a significant preference for a personalised approach to problems, where students may feel anything from workload during university and financial stress to problems with their social network and their personal issues. These issues are a superset of what the charity would like to focus on, which may indicate a need for broadening of scope. The interviews also founded that student would prefer to talk about these mental health issues with a person they can trust, where a logical, yet empathetic solution must be reached for satisfaction. The solution must adhere to these terms.

As for those who may be concerned (the stakeholders), there are many parties to consider. There is an obvious case for the student and the requirements they have, where every student may suffer from a unique problem to the next. Some other stakeholders may involve:

- The charity themselves – these people have a critical role to play, from marketing the app in such a way to ensure success, to committing to maintain and update the app with new features via funding to our software development group.
- Universities and institutions - as an institution, they would want to invest in the wellbeing of the students to see better performance and attainment in assessments, employment and research, further increasing the status and value of the institution.
- Psychologists and healthcare professionals – the app will need to undergo extensive review to determine an optimal final design, agreed upon ideally by an accredited association such as the British Psychological Society (BPS).
- The Government – while it is unclear about any introductory legislation regarding the uses of AI, laws such as the Data Protection Act will need to be upheld.
- Ethics groups – involving the use of personal data may constitute an investigation as to whether the use of this data is safe and secure, and not abused. If proper software engineering principles are not adhered to, there could be resistance in development.

Summary of the Proposed Solution

Upon first hand analysis of the scenario, the brief will take the ideas of the charity and dramatically expand it. From the scenario presented, it is believed the charity truly desires to address as many users as possible. To do this, we must consider not a specific case scenario, but rather the prevalence of many. By the definition of human consciousness, each and every user is identifiably unique, exhibiting truly remarkable exhibits of experience, enthusiasm, and endeavour. Therefore, the solution to the charity's problem must also be unique and special, not to the collective, but to the individual. An extraordinary resolve for such an extraordinary person.

With the analysis of the scenario out of the way, the brief now proposes said unique solution to this complex problem. Our primary proposition, and the basis of the report, will explore the ideas, requirements and feasibility for designing an AI that can provide mental health support in times of need. This can be supported by helper functions that may use the AI to prompt or inform the user. In essence, the AI will have the unique proposition of being a “friend with no judgement”, i.e., the AI will be friendly and sympathetic to the user, so that they can talk about whenever they need, without being afraid of judgement or data breaches. It will also work similarly to the well-established chatbot ChatGPT, by integrating and upgrading its GPT-4 API. Additionally, the AI will be trained on well researched material based on psychology and real-life events, to

ensure accurate and informative feedback is given. This should be similar or better to talking to a real psychologist, but must have the characteristics of a friend or someone that can be relied upon.

This AI would be targeted to the everyone in the student community who desires to use or implement the AI in their day to day lives. To achieve this, the app will be available for students using both android and apple devices. The AI would be available for use anytime, anywhere, so long as the device has an internet connection for the AI to run. Ideally, students would be using the AI to help address hidden and personal problems that they may not feel be comfortable to share with others. Some initial ideas of prompts could be related, but not limited to:

- Loneliness, where the AI can act as a friend to talk to
- Addressing academic and work pressure
- Financial stress and burden, and where to look for help
- Relationship related issues
- Healthy eating and lifestyle
- Suicide prevention strategies

To the user, minimal effort would be required to access the app. A message box would be presented to the user upon launch. Similar to ChatGPT, the user can input their requests relating to any scenarios mentioned and more. The AI would be catered to address these problems by using a user profile gathered from setup or learning from previous prompts, which is stored locally and encrypted to protect the privacy of the user. Using this approach, The AI would then understand the situation and apply critical resolution methods, in a detailed, sophisticated, yet tamed and friendly response (Mrwhosetheboss, 2023).

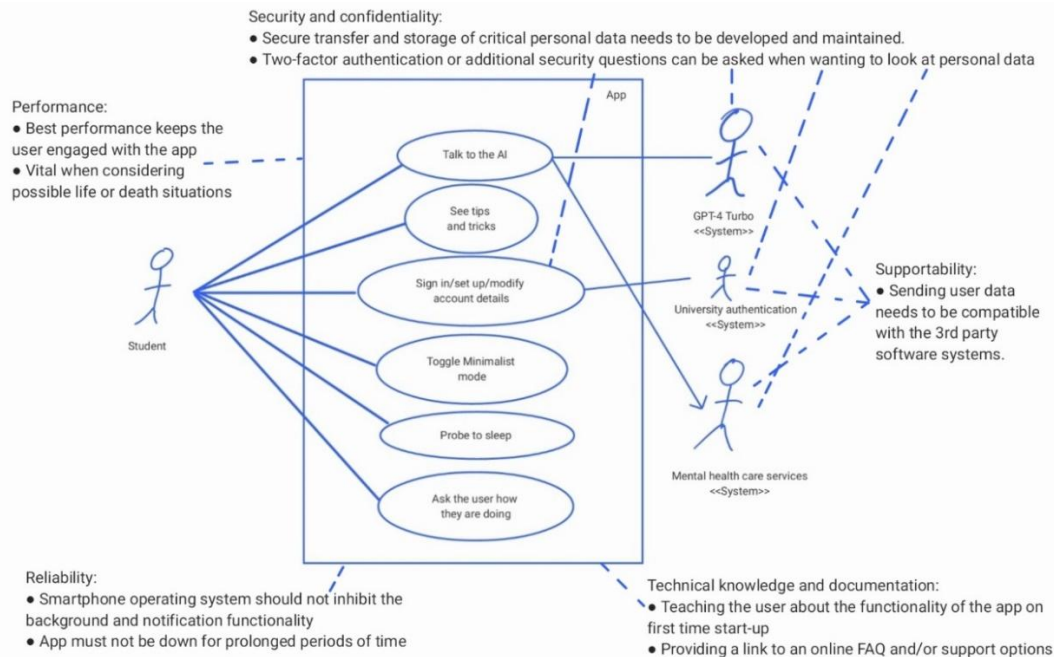
Ideally, the AI would work autonomously to operate the AI and other possible in-app functions, however for the project to succeed, essential architectural achievements are required. These could include creating access to the institutional gateway to allow only registered students to access the app. Another case involves access to the ChatGPT-4 Turbo API to create and run commands on a new ChatGPT with upgraded psychological capabilities and comforting, human-like responses. One other case involves the creation, modification and access of the user profile for use in the AI, and a final case takes into consideration to forward user profiles to mental health services such as BetterHelp or 999, so that the student can be eased when they may need to talk about their problems if the time comes. This is of course optional to respect the privacy of the user.

There are also many minor implementations that can be applied, given the time and budget, via incremental implementation. These may include:

- Functions to Activate a “minimalist mode” which activates “Ultra Power Saving Mode” and “Do Not Disturb” to avoid distractions from other apps.
- Either system or human togglable light and dark mode to provide a more aesthetically pleasing interface during the night.
- Sleep probing, where the app will send a notification if the screen is on outside active hours.
- Adding the suicide prevention hotline if such a conversation appears during an interaction with the AI.
- Tips and tricks tab and notifications generated by the AI, including references to other useful apps
- AI assisted timetable management and deadline scheduling using the university timetable as a reference.

Requirements

Provided below is a use case diagram, which demonstrates a selection of ideas given in the summary with their non-functional requirements. Here the user will have access to many different fields the app could cover (examples covered below), while the system support elements are maintained at their expected requirements.



A brief for each use case is demonstrated, each with its value proposal:

- **Talk to the AI** – The user can ask the AI anything concerning their mental health and how to improve it. It can also be used as a method of venting stress, where the AI will act as a compassionate friend and listen to the user. There can also be integrated support elements built into the AI where, for example, when suicide is mentioned, resources can be shown to the user for support, and information can be sent to these resources for the ease of the user while providing vital contextual information to the mental health worker. Features similar to ChatGPT will also be implemented (the ability to delete chats for example), as well as activating settings for the app when requested.
- **See tips and tricks** – an informative way to provide the user with quick ideas to improve mental health. This can be via the interface of the app, and/or push notifications that may occur whenever the user is active on their phone. These should require little effort, and therefore be easy to implement in the user's life.
- **Sign in/set up/modify account details** – provides the user with easy access to their details, allowing the user to add, edit or delete them as they desire.
- **Toggle minimalist mode** – provides the user with an interface free from distractions, helping the user to focus on themselves or their work.
- **Probe to sleep** – sends a push notification to remind the user to sleep and that sleeping is vital for normal bodily function.
- **Ask the user how they are doing** – sends a push notification to remind the user that they are not alone and do have someone to talk to. Clicking on this immediately loads the app up.

Using the Proposed Solution

In this section, I will formulate a theoretical user story based upon the use case diagram. A user story may look something like this.

A student opens up the app. One of two outcomes is inevitable

Either the user may be using the app for the first time, or re-downloaded the app. In this case, the app will run through the briefing for new users. the app will then request the user to log in to their student account via the gateway access portal. Finally, the app will set up the user's profile, using the user's input. This can include nationality, gender, race or religion to create a personalised experience (Mrwhosetheboss, 2023). The app will then proceed to the main interface.

The second outcome involves the user returning to the app. In this case, the app may request the user to enter login details, depending on the user settings or institutional policy. the app will then proceed to the main interface.

Upon successful sign in, the user may choose to talk to the AI. Algorithms will forward the user input and information to the GPT-4 Turbo server to process the request and send back a response. If the AI feels the user needs additional support, the AI will use affiliate links recommended by the charity and push forward the conversation and user details to that recommendation, provided the user is happy to do so and is applicable. The user may also want to read the tips and tricks. Every one of these is AI created by the app's internal processes, asking for more tips and tricks to GPT-4 Turbo.

Additionally, the user may want to access further functions via the 3 bars on the top right (prototype below). This can lead to the settings option, which could contain the user profile, customisability options, delete personal data and FAQ links, activate minimalist mode (prototype below), or the notification settings, where it can include input for active hours, and what notifications to send.

There are also additional notification functions that act outside the app. These include reminding the user they have someone to talk to if the app has not been launched within a set amount of time. Another function could be to encourage the user to go to sleep. This occurs when the user is active on their phone outside of user set active hours. Finally, a notification could be sent out to provide informative tips and tricks inside the active hours to the user. This should work if the phone is being actively used.

User setup interface:

Please fill in the details below.
Items marked with * must be filled in

*Name:

Date of birth

Gender

Ethnicity

Nationality

[Why do we ask for these details?](#)

Main interface:

Hey there!
Hope you are doing well 😊

Hey Joe, need to talk?

Tap to start talking ➡

Tips and tricks:
Take deep breaths when stressed
< >

Minimalist mode:

100% 🔋

12.04 pm
Wednesday 15th November 2023

Phone Messages Chrome

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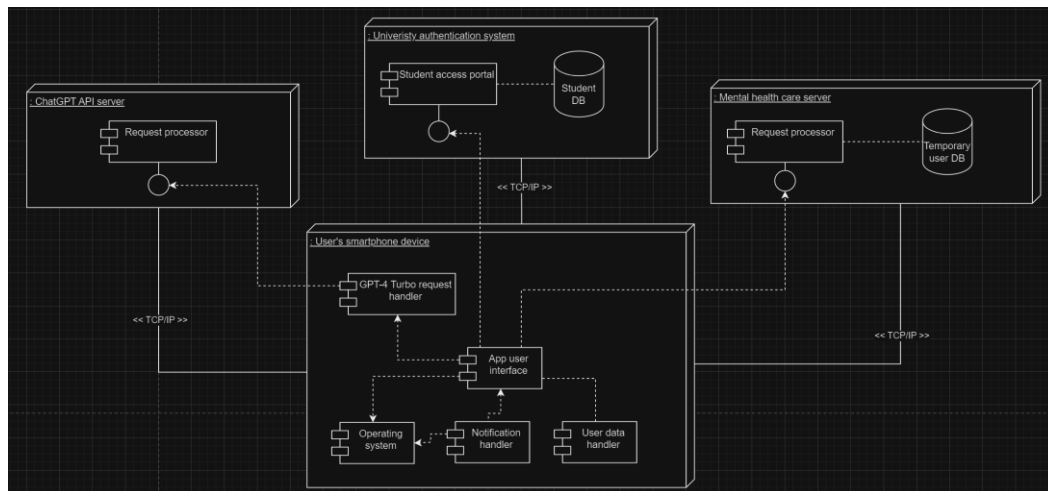
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Turn off minimalist mode

Implementing the Proposed Solution

Below is a provided implementation diagram for the proposed solution. The user phone acts as a primary client, where it makes and handles the requests made to other external servers. There may be three or more external servers depending on how many mental healthcare services are implemented, and the basic architecture may vary for each healthcare server.



In terms of the architecture and platforms, there is actually very little to implement. The only server-side architecture to implement would be for the mental healthcare server to upload the user data and the saved conversation with the AI, so that the agent can better understand the user. This would ideally be handled by the IT department on the healthcare team. The user's smartphone is connected to the internet via a Wi-Fi connection. ChatGPT-4 Turbo's API server already comes with its own in-built interface, so the only requirement here is to build the respective request handler on the client side. OpenAI has documentation on how to create and implement this. Ideally the university authentication system architecture exists, and therefore just requires linking/redirecting from the app's user interface to the portal.

To address the non-functional requirements:

- Performance:
 - Could consider using high performance programming languages such as Kotlin and Swift and the device software development kit.
 - Creating small packets of data that can be sent to each server to be quickly interpreted.
- Security and confidentiality:
 - Encrypting all data sent to and from servers, and data stored on the phone until the app is launched.
 - Using two-factor authentication when wanting to view, edit or delete the details of the user.
- Supportability:
 - Creating documentation regarding the sending and receiving of user data, so that IT companies are able to adhere to the layout and policies.
 - Adhering to documentation created for ChatGPT-4's API and the university portal network.
- Reliability
 - Configuring the whitelist on the smartphone, via the app or the user, so that background processes and notifications are not hindered.
 - Adhering to agile software engineering principles to incrementally deliver the app and any new features or bug fixes.
- Technical knowledge and user documentation
 - Creating online sites, forums and documentation as a guide for users who needs help with any aspect of the app

Conclusion

To summarise, I believe this plan is the soundest solution to what the company is looking for in an app. The idea is very ambitious by covering a wide range of mental health situations, makes use of cutting-edge AI and smartphone technologies, and accounts for the integration for university systems (although not at a full scale as there may be evidence to show that users could operate differently between work and personal life).

The proposal will follow ideal software engineering principles by adhering to disciplined agile development, where, if the need arises, the plan will change to benefit the app's stakeholders. User involvement is encouraged to help isolate more unique and desirable features and improvements. Features will be deployed with robust performance, beautiful user interfaces and extensive testing. The brief recommends no deadline for incremental features, but a deadline for the vital aspects of the app so the core aspect can be pushed to the students for review.

This report has deliberately left some limitations and issues, so that the charity can address these upon further discussion and agreement with stakeholders. Major problems include exceeding the costs of creation and running costs regarding the agile design philosophy and GPT-4 API cost. Institutions could play a key role here by investing in this project, since they would have a significant stake wanting to maintain the mental health of their students. Other considerations involve the frequency of retraining the model, since new data in psychology can prove beneficial to maintaining the AI's cutting-edge aspect, and need for an assertive conclusion for which mental health service(s) could be involved. Candidates could involve the NHS crisis line, BetterHelp and/or Shout.

There are some issues regarding the ethics, legal and sociological aspects of the app. These can include, with their respective solution:

- Cases of racism and discrimination - Approaches could involve modifying the GPT handler, or a more drastic approach could be to remove these fields entirely.
- Addiction to the app or smartphone, leading to a limited social life (Veritasium, 2023) - The app could monitor the user's usage patterns and provide notifications encouraging the user to socially interact with others
- Upholding the data protection act on users – Encryption should be used on all critical personal user data, and data should be stored client-side to reduce liability
- Keeping users comfortable with the use of their data:
 - Giving the users the choice on what data they would like to add, edit or delete
 - Storing user data on their device only to imply control on what they share belongs to them
 - Blatant documentation provided to the user on why their data is collected
- The consideration of training the AI based on user conversations.
 - User data associated with the request becomes pseudonymised
 - Users will have an option to opt in for this feature
- Reliability and accuracy on whether the AI works as expected
 - Needs inspection and accreditation from an independent association such as the BPA
 - Needs feedback from users talking about how to improve the app and the AI's response systems.

References

Veritasium 2023. What The Longest-Running Study on Happiness Reveals. YouTube. [Online]. Available from: <https://www.youtube.com/watch?v=vSQjk9jKarg>.

Mrwhosetheboss 2023. Can AI really save the World? ft. Bill Gates. YouTube. [Online]. Available from: <https://www.youtube.com/watch?v=l9m3IKG8i88>.